VEDANT KALE

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PROFESSIONAL SUMMARY

Final-year B.Tech. CS student and Software Developer Intern targeting Software Development Engineer (SDE) roles in backend engineering and AI systems. Built RAG and GraphRAG pipelines that raised Recall@10 from 42.6% to 73.3%. Skilled in C++, Python, Java, and DSA; 150+ LeetCode problems solved.

EDUCATION

Pimpri Chinchwad University (PCU), Pune, India **Aug 2023 – May 2027**
B.Tech. in Computer Science & Engineering *CGPA: 7.69 / 10.0*

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL

Frameworks & Web: FastAPI, Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, Microservices

Databases & Retrieval: PostgreSQL, MongoDB, Redis, Neo4j, Pinecone, pgvector, SQLAlchemy, BM25, Hybrid Search, Cross-Encoder Reranking

AI and ML: RAG, GraphRAG, Knowledge Graphs, Embeddings, RAGAS, LangChain, LangGraph, Ollama, Cohere Rerank, PyTorch, Scikit-learn, Pandas, NumPy

Cloud, DevOps & Testing: Docker, Git, Linux, CI/CD, AWS (EC2, S3, Lambda, DynamoDB), Pytest, Automated Testing, Latency Monitoring

Core CS: Data Structures and Algorithms (DSA), OOP, Operating Systems, DBMS, System Design, Distributed Systems, Software Architecture

WORK EXPERIENCE

Software Developer Intern — Karst Technologies Private Limited, Pune, India **Jul 2026 – Sep 2026**
Technologies: JavaScript, Node.js, Express.js, React.js, MongoDB, REST APIs, Git

- Developed and maintained 3 application components, resolving 5 software defects across development workflows.
- Integrated and tested 4 REST API endpoints with database operations, validating data flow across components.
- Executed 10 functional test cases, documenting defects and verifying application behavior before feature delivery.

PROJECTS

Neuro-Adaptive GraphRAG Platform **Jan 2026 – Apr 2026**
Python, FastAPI, Neo4j, Pinecone, Cohere, RAGAS | github.com/VedantKale244/RAG-pipeline *Personal Project*

- Built a hybrid GraphRAG pipeline (Neo4j, Pinecone, Cohere reranking) with 534 passing tests across 36 modules.
- Implemented RAGAS evaluation gates for retrieval quality and faithfulness with a 2.5 s p95 latency threshold.
- Developed a self-optimization framework (quality 50%, latency 30%, feedback 20%) with 15% canary rollout.

ArthaSEC – Financial RAG and Evidence Verification Engine **May 2026 – Jul 2026**
Python, FastAPI, BM25, Hybrid Search, SQLAlchemy | github.com/VedantKale244/ArthaSEC *Personal Project*

- Built a financial retrieval engine over 352 SEC filings (32,000 chunks), raising Recall@10 from 42.6% to 73.3%.
- Added citation verification and deterministic calculations to validate financial metrics and flag unsupported figures.
- Optimized retrieval with BGE cross-encoder reranking, reaching about 17 ms p50 latency in fast mode.

Adaptive Two-Tier Malware Detection **Ongoing Research**
Python, Machine Learning, Random Forest, Evidential Deep Learning, API Security *Research Project*

- Designed a two-tier detection framework (Random Forest + sandbox) for SQLi, XSS, malware uploads, and RCE on 9,000 API requests.
- Added confidence-based escalation and feedback retraining: 98.1% accuracy, 58.8% fewer deep inspections.
- Evaluated on 10,000 CIC-IDS/EMBER records, reaching 99.30% accuracy and 0.003 ms CPU inference latency.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Data Structures and Algorithms Specialization – Amazon (via Coursera), 2025; The Complete Full-Stack Web Development Bootcamp – Udemy, 2026; Software Design and Architecture Specialization – Coursera (In Progress).
- **Competitive Programming:** LeetCode rating 1,480 (150+ solved); CodeChef rating 1,824; CodeChef DSA rating 1,659.